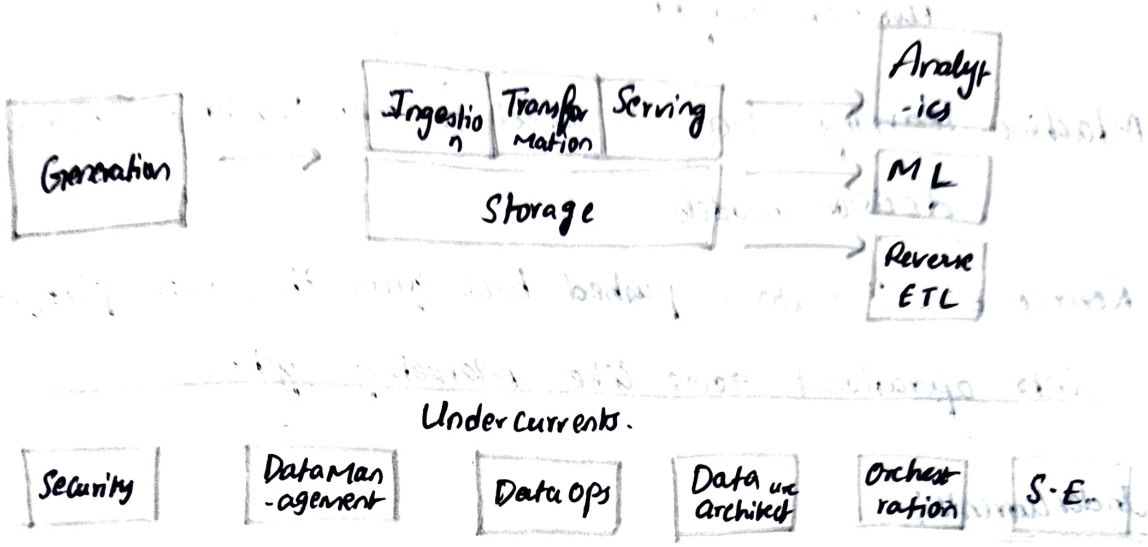
Data Engineering Lifecycle



Explanation of Each step :-

* Generation :- Data is created by source systems such as applications, sensors, logs or user interactions.

* Ingestion : Raw data is collected and moved from source systems into data platform.

* Storage :- Data is stored in data lakes, warehouses or databases.

* Transformation :- Data is cleaned, validated, reshaped into usable formats.

* Serving :- Processed data is exposed for consumption by users and systems.

Data Consumers :-

Analytics :- Data is visualized for reporting, dashboards and business insights.

Machine learning :- Data is used to train, Validate and run decision models.

Reverse ETL :- Data is pushed back from the data platform into operational tools like Marketing systems.

Undercurrents :-

* Security :- Ensures data privacy, accen Control, encryption etc
* Data Mangement :- Handles data quality.
* DataOps :- Applies testing, monitoring, realibility practices to data pipelines.
* Data architecture :- Defines how data System, tools and flows are Structured and integrated.
* Orchestration :- Coordinates and Schedules data workflows.
* Software Engineering :- Applies coding practices. Such as version control, testing etc.